

Bronze knife

c. 600–200 BCE

Surgical knives were used in Ancient Egypt—an early center of medical excellence. These knives might also have been used to remove organs prior to mummification.



curved blade

supporting brace

Bone saw
16th century

In Europe, early amputation saws for cutting through bone were used without anesthetic. Patients were given only alcohol to alleviate pain. Many saws had ornate handles, which served only to harbour more germs.

cutting edge



serrated cutting blade

Amputation knife
18th century

Before sawing through the bone, surgeons used the inner edge of a curved knife blade to make a round cut through the skin and muscle.

SURGERY

FROM ANCIENT TIMES, SURGERY HAS BEEN USED TO INVESTIGATE AND TREAT DISEASE OR INJURY

As the most invasive kind of medical procedure, surgery is often literally a matter of life or death. But operations that would have been considered risky 100 years ago are today a matter of routine.

Surgery is fraught with three risks: pain, blood loss, and infection. The history of surgery is largely the story of how science has been able to mitigate these risks. In the 20th century, improved anesthetics and storage and transfusion of compatible blood types meant more people survived operations. Germ theory and effective antiseptics led to a drop in infection rates.

Blood bag
1950s

In the early 1900s, it was discovered that a substance called citrate could stop blood from coagulating. This made possible the storing of blood and blood products for routine surgery and emergencies.

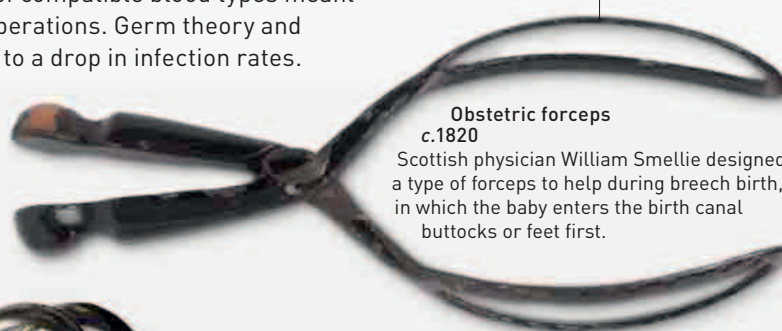


label with blood type information

blade to be positioned around head of baby

Obstetric forceps
c. 1820

Scottish physician William Smellie designed a type of forceps to help during breech birth, in which the baby enters the birth canal buttocks or feet first.

**Tonsil guillotine**
1850s

Removing tonsils was a popular way of treating ongoing throat infections until the 1950s. It fell out of favor as knowledge of infection changed.



button to move blades

cutting blade

heated bulb used to cauterize skin surface

Petit tourniquet
18th century

In 1718, French surgeon Louis Petit developed a screw-type device known as a tourniquet to tighten a limb strap and stop blood flow.



screw for tightening